

Module 7: Introduction to Gibbs Sampling

Rebecca C. Steorts

Agenda

In this lab, we will deriving conditional distributions, code a Gibbs sampler, and analyze the output of the Gibbs sampler.

Problem Statement

Consider the following Exponential model for observation(s)
 $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_n)$.¹:

$$p(x|a, b) = ab \exp(-abx)I(x > 0),$$

where the x_i are assumed to be iid for $i = 1, \dots, n$. and suppose the prior is

$$p(a, b) = \exp(-a - b)I(a, b > 0).$$

You want to sample from the posterior $p(a, b|x_{1:n})$. You may assume that

$$a_0 = 0.25, b_0 = 0.25$$

when coding up your Gibbs sampler.

¹The data can be found in data-exponential.csv.

Tasks

1. Find the conditional distributions needed for implementing a Gibbs sampler.
2. Code up your own Gibbs sampler using part (1).
3. Run the Gibbs sampler, providing convergence diagnostics.
4. Plot a histogram or a density estimate of the estimated posterior using (2) and (3).
5. How do you know that your estimated posterior in (3) is reliable?

Task 1:

Consider the following Exponential model for observation(s)
 $x = (x_1, \dots, x_n)$.²:

$$p(x|a, b) = ab \exp(-abx)I(x > 0)$$

and suppose the prior is

$$p(a, b) = \exp(-a - b)I(a, b > 0).$$

You want to sample from the posterior $p(a, b|x)$.

²Please note that in the attached data there are 40 observations, which can be found in data-exponential.csv.

Task 1: Conditional distributions

$$\begin{aligned} p(\mathbf{x}|a, b) &= \prod_{i=1}^n p(x_i|a, b) \\ &= \prod_{i=1}^n ab \exp(-abx_i) \\ &= (ab)^n \exp\left(-ab \sum_{i=1}^n x_i\right). \end{aligned}$$

The function is symmetric for a and b , so we only need to derive $p(a|\mathbf{x}, b)$.

Task 1: Conditional distributions

This conditional distribution satisfies

$$\begin{aligned} p(a|\mathbf{x}, b) &\propto_a p(a, b, \mathbf{x}) \\ &= p(\mathbf{x}|a, b)p(a, b) \\ &= \text{fill in full details for lab this week} \end{aligned}$$

Task 2: Gibbs sampling code

```
knitr::opts_chunk$set(cache=TRUE)
library(MASS)
data <- read.csv("data-exponential.csv", header = FALSE)
```


Task 2: Gibbs sampling code

```
#####  
# This function is a Gibbs sampler  
#  
# Args  
#   start.a: initial value for a  
#   start.b: initial value for b  
#   n.sims: number of iterations to run  
#   data: observed data, should be in a  
#         # data frame with one column  
#  
# Returns:  
#   A two column matrix with samples  
#     # for a in first column and  
# samples for b in second column  
#####
```

Task 2: Gibbs sampling code

```
sampleGibbs <- function(start.a, start.b, n.sims, data){  
  # get sum, which is sufficient statistic  
  x <- sum(data)  
  # get n  
  n <- nrow(data)  
  # create empty matrix, allocate memory for efficiency  
  res <- matrix(NA, nrow = n.sims, ncol = 2)  
  res[1,] <- c(start.a, start.b)  
  for (i in 2:n.sims){  
    # sample the values  
    res[i,1] <- rgamma(1, shape = n+1,  
                      rate = res[i-1,2]*x+1)  
    res[i,2] <- rgamma(1, shape = n+1,  
                      rate = res[i,1]*x+1)  
  }  
  return(res)  
}
```

Task 3: Run the Gibbs sampler

```
# run Gibbs sampler  
n.sims <- 10000  
# return the result (res)  
res <- sampleGibbs(.25,.25,n.sims,data)  
head(res)
```

```
##           [,1]      [,2]  
## [1,] 0.250000 0.250000  
## [2,] 2.578242 0.1556585  
## [3,] 3.555271 0.1607362  
## [4,] 2.718913 0.1913754  
## [5,] 2.303847 0.2090047  
## [6,] 1.767024 0.2161758
```

Task 4

Plot a histogram or a density estimate of the estimated posterior using tasks (2) and (3).

Finish this for homework.

Task 5

How do you know that your estimated posterior in task (3) is reliable?

Finish for homework.